



SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO1-AT3-Coding Challenge

View Only

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QUESTIONS

Q1

Smart University Payroll Management System
25 marks



Q2

Smart Vehicle Insurance Claim System
25 marks



2/2 answered

Q1 Smart University Payroll Management System

25 marks

A university employs three categories of staff: Teaching Faculty, Technical Staff, and Visiting Faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements Create a superclass Employee with the following private data members: employeeId employeeName department basicSalary Provide: Default constructor Parameterized constructor Getter and Setter methods displayEmployee() calculateSalary() Create the following subclasses: TeachingFaculty teachingHours incentivePerHour Salary: $\text{basicSalary} + (\text{teachingHours} \times \text{incentivePerHour})$ TechnicalStaff overtimeHours overtimeRate Salary: $\text{basicSalary} + (\text{overtimeHours} \times \text{overtimeRate})$ VisitingFaculty lecturesHandled paymentPerLecture Salary: $\text{lecturesHandled} \times \text{paymentPerLecture}$ Override calculateSalary() in all subclasses. Create an array of Employee objects and demonstrate runtime polymorphism. Display: Employee details Monthly salary Highest-paid employee Average salary Search an employee using the employee ID.

Test Case 1 (Normal Case)

Input

3

TeachingFaculty

101

Dr. Arun

CSE

50000

20

500

TechnicalStaff

102

Mr. Ravi

ECE

35000

15

300

VisitingFaculty

103

Dr. Kumar

IT

0

40

1200

Search Employee ID:

102

Expected Output

Employee Details

Employee ID : 101

Name : Dr. Arun

Department : CSE

Monthly Salary : 60000.0

Employee ID : 102

Name : Mr. Ravi

Department : ECE

Monthly Salary : 39500.0

Employee ID : 103

Name : Dr. Kumar

Department : IT

Monthly Salary : 48000.0

Highest Paid Employee

Employee ID : 101

Salary : 60000.0

Average Salary

49166.67

Employee Found

Employee ID : 102

Name : Mr. Ravi

Salary : 39500.0

Your Code

```
1 import java.util.*;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int n =sc.nextInt();
```

```
6      Employee e[] = new Employee[n];
7      for(int i=0;i<n;i++){
8          String type=sc.next();
9          int id=sc.nextInt();
10         sc.nextLine();
11         String name=sc.nextLine();
12         String dept=sc.nextLine();
13         double basic=sc.nextDouble();
14         if(type.equals("TeachingFaculty"))
15             e[i]=new TeachingFaculty(id,name,dept,basic,sc.nextInt()
16         else if(type.equals("TechnicalStaff"))
17             e[i]=new TechnicalStaff(id,name,dept,basic,sc.nextInt()
18         else
19             e[i]= new VisitingFaculty(id,name,dept,basic,sc.nextInt
20     }
21     int search=sc.nextInt();
22     Employee high=e[0],found=null;
23     double sum=0;
24     System.out.println("Employee Details");
25     for(Employee x:e){
26         System.out.println();
27         System.out.println("Employee ID : "+x.id);
28         System.out.println("Name : "+x.name);
29         System.out.println("Department : "+x.dept);
30         System.out.println("Monthly Salary : "+x.salary());
31         sum+=x.salary();
32         if(x.salary()>high.salary())
33             high=x;
34         if(x.id==search)
35             found=x;
36     }
37     System.out.println();
38     System.out.println("Highest Paid Employee");
39     System.out.println("Employee ID : "+high.id);
```

```
40     System.out.println("Salary : "+high.salary());
41     System.out.println();
42     System.out.println("Average Salary");
43     System.out.printf("%.2f%n", sum/n);
44     if(found != null){
45         System.out.println();
46         System.out.println("Employee Found");
47         System.out.println("Employee ID : "+found.id);
48         System.out.println("Name : "+found.name);
49         System.out.println("Salary : "+found.salary());
50     }
51     sc.close();
52 }
53 }
54 abstract class Employee{
55     int id;
56     String name,dept;
57     double basic;
58     Employee(int id,String name,String dept,double basic){
59         this.id=id;
60         this.name=name;
61         this.dept=dept;
62         this.basic=basic;
63     }
64     abstract double salary();
65 }
66 class TeachingFaculty extends Employee{
67     int hours;
68     double rate;
69     TeachingFaculty(int id,String name,String dept,double basic,int h,d
70         super(id,name,dept,basic);
71         hours=h;
72         rate=r;
73 }
```

```
74     double salary() {
75         return basic+hours*rate;
76     }
77 }
78 class TechnicalStaff extends Employee{
79     int hours;
80     double rate;
81     TechnicalStaff(int id,String name,String dept,double basic,int h,double
82         super(id,name,dept,basic);
83         hours=h;
84         rate=r;
85     }
86     double salary() {
87         return basic+hours*rate;
88     }
89 }
90 class VisitingFaculty extends Employee{
91     int lectures;
92     double payment;
93     VisitingFaculty(int id,String name,String dept,double basic,int l,double d
94         super(id,name,dept,basic);
95         lectures=l;
96         payment=p;
97     }
98     double salary() {
99         return lectures*payment;
100     }
101 }
```



Input

3
TeachingFaculty
101
Dr.Arun

Output

```
Employee Details  
  
Employee ID : 101  
Name : Dr.Arun  
Department : CSE  
Monthly Salary : 60000.0
```

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